

Fallout 4 to Fallout 76 Textures

Requirements

1. Python 3.x
2. Pillow
 - a. Used for image manipulation
3. numpy
 - a. Used for pixel math
4. tkinter
 - a. Used to create the interface
5. [texconv.exe](#)
 - a. Used to convert the images to the correct format

Instructions

Initial Setup

Follow the instructions below to get your environment set up:

1. Install python
2. Run the following command once python is installed:
 - a. `pip install pillow numpy`
 - i. tkinter usually comes with python, but if not, install it
3. Install texconv.exe either from the [GitHub link](#) or by using the below command in a command prompt or PowerShell
 - a. `winget install Microsoft.DirectXTex.Texconv`

Running the file

Now that you have the requirements installed, you can start using the program. To get started:

1. Move all of your Fallout 4 texture files to a folder. Each folder should only contain the textures for 1 material.
 - a. Make sure your dds files use the following naming convention:
 - i. Normal: `_n.dds`
 - ii. Specular: `_s.dds`
 - iii. Diffuse: `_d.dds`

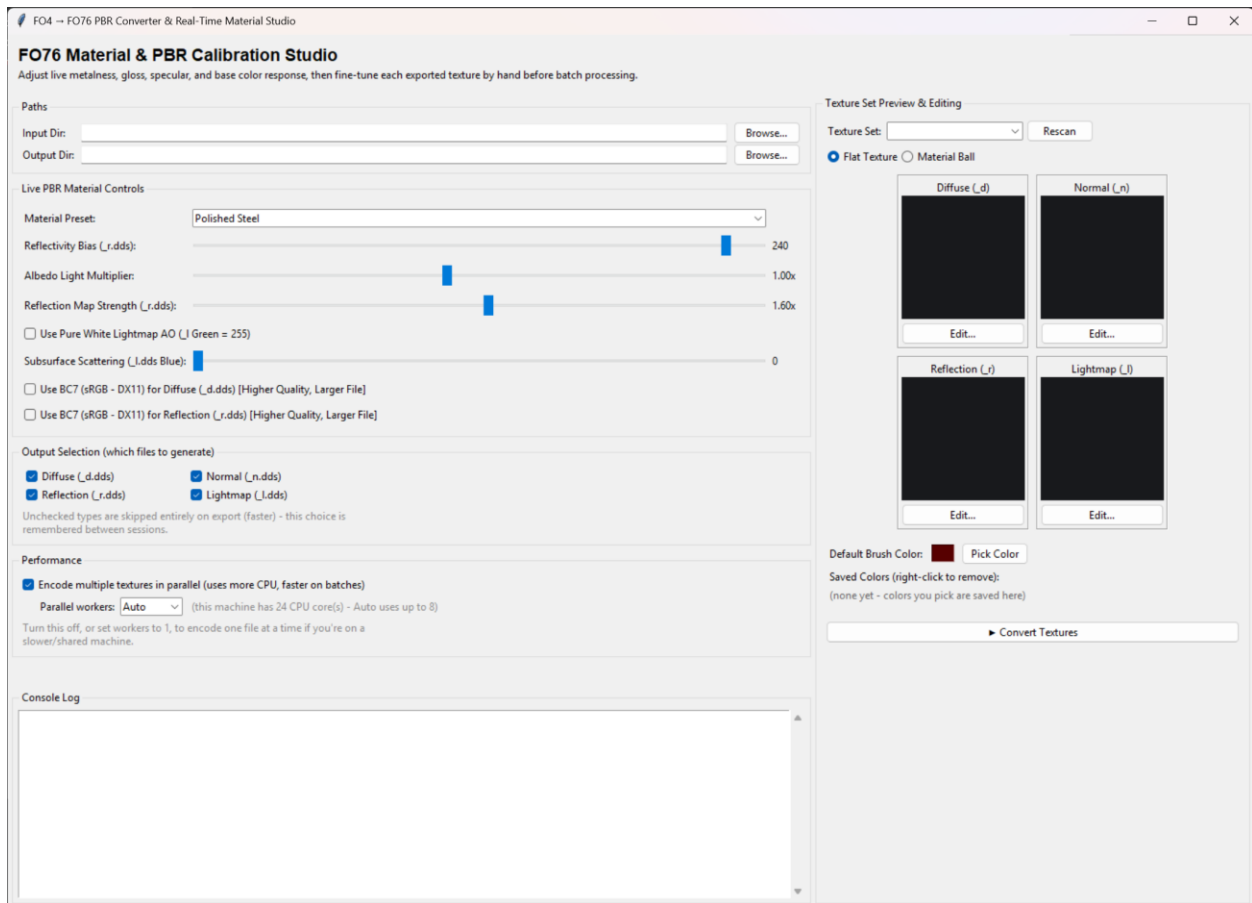
2. Run the python file using the following command:

- a. `python "C:\Path\To\Downloaded\File\Convert f04 textures to f76 textures.py"`
 - i. **Note:** make sure to replace the script above with the path to the file

Note: To make things easier, you can also run this from VS Code, or create a bat file

Converting Textures

Once the file is running, you should see the interface below:



The different elements of the UI work to create the different texture files for Fallout 76:

- Paths
 - This section allows you to choose your input/output directory
 - Select the input directory/folder where your FO4 textures are stored
 - Select the output directory/folder where you want your FO76 textures to be stored
- Live PBR Material Controls
 - Material Preset – Automatically adjusts the sliders to help with the reflection map

- Reflectivity Bias – controls how reflective we want the material to be, it shifts the pixels on the _r map to be brighter or darker
- Albedo Light Multiplier – adjusts the color of the diffuse map making it brighter or darker depending on your preference
 - 1.0 – keeps it the same as the original
- Reflection map strength – controls how strongly the difference between the diffuse map and the specular map’s red channel gets amplified
 - Basically controls the shininess
- Use Pure White Lightmap AO – doesn’t attempt to create an AO, just creates a blanket map
- Subsurface Scattering – use if you need some subsurface scattering
- Use BC7 for Diffuse – uses the higher quality file format for the diffuse map
- Use BC7 for the Reflection – use the higher quality file format for the diffuse map
- Output Selection
 - Check the boxes for each file you wish to generate
- Performance
 - Check the box in this section to generate the texture all at the same time
 - You can also choose how many CPU cores to run on using the “Parallel workers” option
- Texture Set Preview and Editing
 - You can choose between a PBR Sphere (Material Ball) or the Flat Textures
 - This section auto builds a preview of the 4 textures and allows for some editing
 - The interface is not perfect, so, for better control, more editing may be required outside of the program.
 - You can also save custom colors
 - Right click on each saved color to remove it

When you are ready to convert your textures, hit the “Convert Textures” button and they will automatically be built and saved to your selected location.

Final Notes:

This script isn’t perfect, but it might help speed things up. It also doesn’t account for environment maps, so you still may need to manually adjust some files.